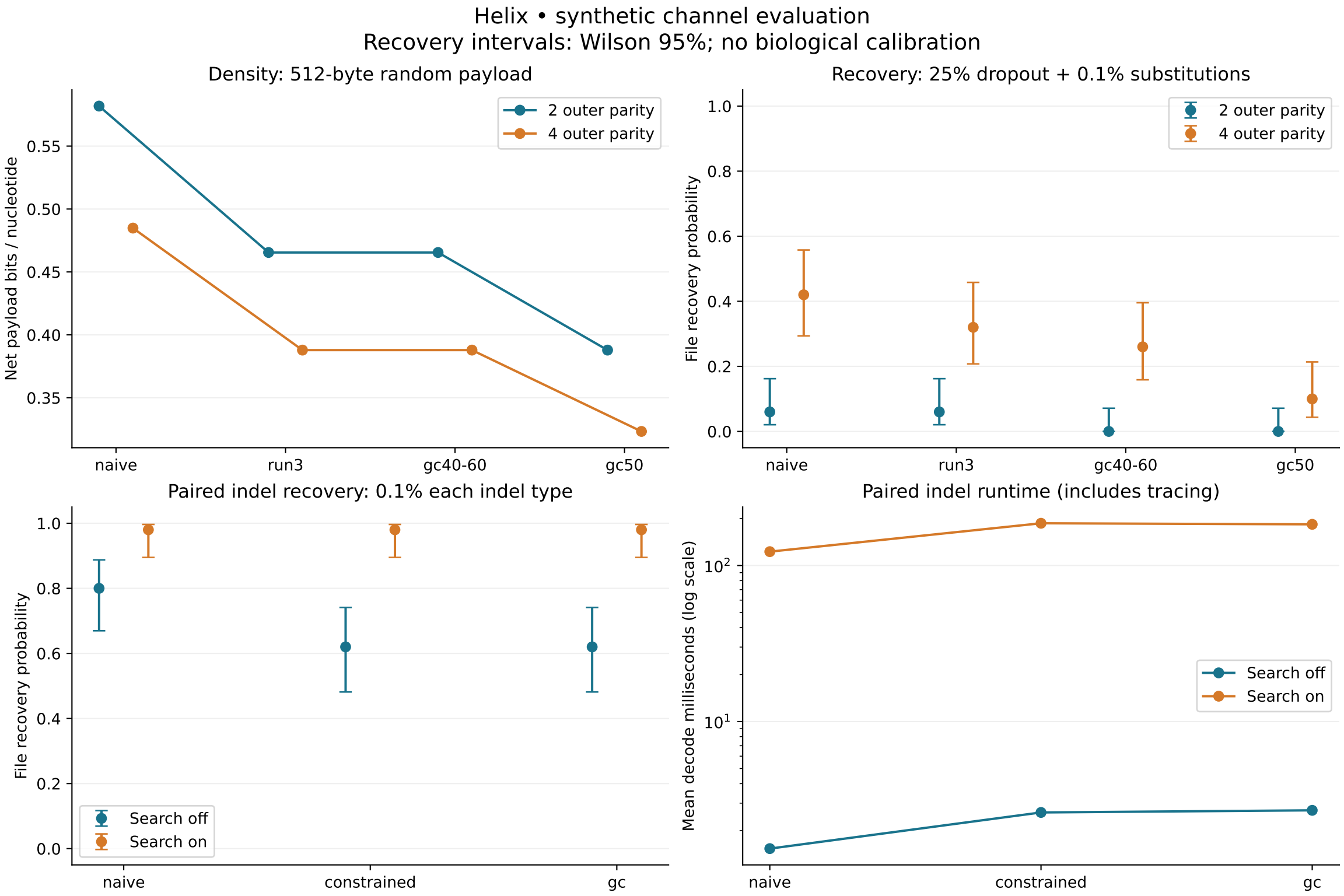
Helix • synthetic channel evaluation
Recovery intervals: Wilson 95%; no biological calibration
Density: 512-byte random payload
Net payload bits / nucleotide
0.55
0.50
0.45
0.40
0.35
2 outer parity
4 outer parity
naive
run3
gc40-60
gc50
Recovery: 25% dropout + 0.1% substitutions
File recovery probability
1.0
0.8
0.6
0.4
0.2
0.0
2 outer parity
4 outer parity
naive
run3
gc40-60
gc50
Paired indel recovery: 0.1% each indel type
File recovery probability
1.0
0.8
0.6
0.4
0.2
0.0
Search off
Search on
naive
constrained
gc
Paired indel runtime (includes tracing)
Mean decode milliseconds (log scale)
10^2
10^1
Search off
Search on
naive
constrained
gc